

Instructions for Extended Match Questions

Use this slide to understand how to answer an EMQ

Instructions:

1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. Please click [here](#) to view a sample template file for answering.
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

Consider the following matrices: $R_1 = \begin{bmatrix} 1 & 0 & \frac{5}{2} & 1 \\ 0 & 1 & \frac{3}{4} & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $R_2 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $R_3 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$,

$$R_4 = \begin{bmatrix} 1 & 0 & \frac{5}{2} & 1 \\ 0 & 1 & \frac{3}{4} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

List of Options:

1. R_1 is a row echelon form of A .
2. R_1 is the reduced row echelon form of A
3. R_2 is the reduced row echelon form of A
4. R_3 is the reduced row echelon form of A
5. R_4 is the reduced row echelon form of A
6. $\text{Nullspace}(A) = \{(-t, -t, 0, t) \mid t \in \mathbb{R}\}$
7. $\text{Nullspace}(A) = \{(-t, \frac{t}{2}, t, 0) \mid t \in \mathbb{R}\}$
8. $\text{Nullspace}(A) = \{(-\frac{5}{2}t_1 - t_2, -\frac{3}{4}t_1 - t_2, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$
9. $\text{nullity}(A) = 1$
10. $\text{nullity}(A) = 2$
11. Basis of $\text{Nullspace}(A) = \{(-\frac{5}{2}, -\frac{3}{4}, 1, 0), (-1, -1, 0, 1)\}$
12. Basis of $\text{Nullspace}(A) = \{(1, -\frac{1}{2}, -1, 0)\}$
13. Basis of $\text{Nullspace}(A) = \{(1, 1, 0, -1)\}$

List of Questions:

Q1 Consider the matrix $A = \begin{bmatrix} 2 & 0 & 5 & 2 \\ 1 & 2 & 4 & 3 \\ 2 & 0 & 7 & 2 \end{bmatrix}$, which of the above options are true for A ?

Q2 Consider the matrix $A = \begin{bmatrix} 2 & 0 & 5 & 2 \\ 1 & 2 & 4 & 3 \\ 3 & 2 & 9 & 5 \end{bmatrix}$, which of the above options are true for A ?

Q3 Consider the matrix $A = \begin{bmatrix} 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 \\ 2 & 0 & 2 & 1 \end{bmatrix}$, which of the above options are true for A ?